Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

4.2 Practice Questions

Solve the problem and pick the best answer choice given.

Numbers: **All numbers used are real numbers.**

Figures: **A figure in a Quantitative Reasoning question gives information useful in solving the problem. Figures are drawn as accurately as possible, except as noted. Lines shown as straight are straight. Lines that look jagged may also be straight. Points, angles, regions, etc., are positioned as shown. All figures are flat unless otherwise noted.**

Questions 1 to 82 — Difficulty: Easy

1. Working at a constant rate, a copy machine makes 20 copies of a one-page document per minute. If the machine works at this constant rate, how many hours does it take to make 4,800 copies of a one-page document?

   A. 4

   B. 5

   C. 6

   D. 7

   E. 8

2. If $x + y = 2$ and $x^2 + y^2 = 2$, what is the value of xy?

   A. –2

   B. –1

   C. 0

   D. 1

   E. 2

3. The sum S of the first n consecutive positive even integers is given by $S = n(n + 1)$. For what value of n is this sum equal to 110?

   A. 10

   B. 11

   C. 12

   D. 13

   E. 14

4. 6(87.30 + 0.65) − 5(87.30) =

   A. 3.90

   B. 39.00

   C. 90.90

   D. 91.20

   E. 91.85

5. What is the value of $x^2yz - xyz^2$, if $x = -2$, $y = 1$, and $z = 3$?

   A. 20

   B. 24

   C. 30

   D. 32

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. 48

6. If T is a set of 35 consecutive integers, of which 17 are negative, what is the sum of all the integers in T?

A. –2

B. –1

C. 0

D. 1

E. 2

7. If $x > y$ and $y > z$, which of the following represents the greatest number?

A. $x - z$

B. $x - y$

C. $y - x$

D. $z - y$

E. $z - x$

8. 38, 69, 22, 73, 31, 47, 13, 82

Which of the following numbers is greater than three-fourths of the numbers but less than one-fourth of the numbers in the list above?

A. 56

B. 68

C. 69

D. 71

E. 73

9. A rug manufacturer produces rugs at a cost of $75 per rug. What is the manufacturer's gross profit from the sale of 150 rugs if $\frac{2}{3}$ of the rugs are sold for $150 per rug and the rest are sold for $200 per rug?

A. $10,350

B. $11,250

C. $13,750

D. $16,250

E. $17,800

10. The value of Maureen's investment portfolio has decreased by 5.8 percent since her initial investment in the portfolio. If her initial investment was $16,800, what is the current value of the portfolio?

A. $7,056.00

B. $14,280.00

C. $15,825.60

D. $16,702.56

E. $17,774.40

11. Company C produces toy trucks at a cost of $5.00 each for the first 100 trucks and $3.50 for each additional truck. If 500 toy trucks were produced by Company C and sold for $10.00 each, what was Company C's gross profit?

A. $2,250

B. $2,500

C. $3,100

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

C. $3,100

D. $3,250

E. $3,500

Division	Profit or Loss (in millions of dollars)				
	1991	1992	1993	1994	1995
A	1.1	(3.4)	1.9	2.0	0.6
B	(2.3)	5.5	(4.5)	3.9	(2.9)
C	10.0	(6.6)	5.3	1.1	(3.0)

12. The annual profit or loss for the three divisions of Company T for the years 1991 through 1995 are summarized in the table shown, where losses are enclosed in parentheses. For which division and which three consecutive years shown was the division's profit or loss for the three-year period closest to $0?

    A. Division A for 1991–1993

    B. Division A for 1992–1994

    C. Division B for 1991–1993

    D. Division B for 1993–1995

    E. Division C for 1992–1994

13. A computer appliance dealer bought a computer at a wholesale price of $500 and then sold it at a 20 percent discount off the suggested retail price. If the dealer made a 12 percent profit on the wholesale price, what was the suggested retail price of the computer?

    A. $560

    B. $580

    C. $660

    D. $700

    E. $730

14. Pat invested x dollars in a fund that paid 8 percent annual interest, compounded annually. Which of the following represents the value, in dollars, of Pat's investment plus interest at the end of 5 years?

    A. $5(0.08x)$

    B. $5(1.08x)$

    C. $[1 + 5(0.08)]x$

    D. $(1.08)^5 x$

    E. $(1.08x)^5$

15. There are five sales agents in a certain real estate office. One month Andy sold twice as many properties as Ellen, Bob sold 3 more than Ellen, Cary sold twice as many as Bob, and Dora sold as many as Bob and Ellen together. Who sold the most properties that month?

    A. Andy

    B. Bob

    C. Cary

    D. Dora

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. Ellen

16. In a field day at a school, each child who competed in n events and scored a total of p points was given an overall score of $\frac{p}{n} + n$. Andrew competed in 1 event and scored 9 points. Jason competed in 3 events and scored 5, 6, and 7 points, respectively. What was the ratio of Andrew's overall score to Jason's overall score?

A. $\frac{10}{23}$

B. $\frac{7}{10}$

C. $\frac{4}{5}$

D. $\frac{10}{9}$

E. $\frac{12}{7}$

17. A certain work plan for September requires that a work team, working every day, produce an average of 200 items per day. For the first half of the month, the team produced an average of 150 items per day. How many items per day must the team average during the second half of the month if it is to attain the average daily production rate required by the work plan?

A. 225

B. 250

C. 275

D. 300

E. 350

18. A company sells radios for $15.00 each. It costs the company $14.00 per radio to produce 1,000 radios and $13.50 per radio to produce 2,000 radios. How much greater will the company's gross profit be from the production and sale of 2,000 radios than from the production and sale of 1,000 radios?

A. $500

B. $1,000

C. $1,500

D. $2,000

E. $2,500

19. Working at their respective constant rates, machine A makes 100 copies in 12 minutes and machine B makes 150 copies in 10 minutes. If these machines work simultaneously at their respective rates for 30 minutes, what is the total number of copies that they will produce?

A. 250

B. 425

C. 675

D. 700

E. 750

20. Company X mailed out a questionnaire. Sixty percent of those who received the questionnaire by mail responded. Company X needed 300 responses. What is the minimum number of questionnaires that Company X should have mailed in order to get the necessary responses?

A. 400

B. 420

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

C. 480

D. 500

E. 600

21. The toll T, in dollars, for a truck using a certain bridge is given by the formula $T = 1.50 + 0.50(x - 2)$, where x is the number of axles on the truck. What is the toll for an 18-wheel truck that has 2 wheels on its front axle and 4 wheels on each of its other axles?

A. $2.50

B. $3.00

C. $3.50

D. $4.00

E. $5.00

22. For what value of x between –4 and 4, inclusive, is the value of $x^2 - 10x + 16$ the greatest?

A. –4

B. –2

C. 0

D. 2

E. 4

23. If $x = -\frac{5}{8}$ and $y = -\frac{1}{2}$, what is the value of the expression $-2x - y^2$?

A. $-\frac{3}{2}$

B. –1

C. 1

D. $\frac{3}{2}$

E. $\frac{7}{4}$

24. At the opening of a trading day, stock X had a value of $40.00 per share and stock Y had a value of $25.00 per share. At the closing of the trading day, stock X had a value of $41.00 per share and stock Y had a value of $28.50 per share. For this trading day, the percent increase in the value of stock Y was how many percentage points greater than the percent increase in the value of stock X?

A. 2.5

B. 6.25

C. 10

D. 11.5

E. 12.5

25. For positive integers a and b, the remainder when a is divided by b is equal to the remainder when b is divided by a. Which of the following could be a value of ab?

I. 24

II. 30

III. 36

A. II only

B. III only

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. III only

C. I and II only

D. II and III only

E. I, II, and III

26. List S consists of the positive integers that are multiples of 9 and are less than 100. What is the median of the integers in S?

A. 36

B. 45

C. 49

D. 54

E. 63

27. A rope 20.6 meters long is cut into two pieces. If the length of one piece of rope is 2.8 meters shorter than the length of the other, what is the length, in meters, of the longer piece of rope?

A. 7.5

B. 8.9

C. 9.9

D. 10.3

E. 11.7

28. If x and y are integers and $x - y$ is odd, which of the following must be true?

I. xy is even.

II. $x^2 + y^2$ is odd.

III. $(x + y)^2$ is even.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

29. On Monday, the opening price of a certain stock was $100 per share and its closing price was $110 per share. On Tuesday the closing price of the stock was 10 percent less than its closing price on Monday, and on Wednesday the closing price of the stock was 4 percent greater than its closing price on Tuesday. What was the approximate percent change in the price of the stock from its opening price on Monday to its closing price on Wednesday?

A. A decrease of 6%

B. A decrease of 4%

C. A decrease of 1%

D. An increase of 3%

E. An increase of 4%

30. 1 − 0.000001 =

A. (1.01)(0.99)

B. (1.11)(0.99)

C. (1.001)(0.999)

D. (1.111)(0.999)

E. (1.0101)(0.0909)

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

31. In a certain history class of 17 juniors and seniors, each junior has written 2 book reports and each senior has written 3 book reports. If the 17 students have written a total of 44 book reports, how many juniors are in the class?

A. 7

B. 8

C. 9

D. 10

E. 11

32. $|-4|(|-20| - |5|) =$

A. −100

B. −60

C. 60

D. 75

E. 100

33. What is the greatest integer k for which 32.45×10^k is less than 1?

A. –2

B. 1

C. 0

D. 1

E. 2

34. How many integers x satisfy both $2 < x \leq 4$ and $0 \leq x \leq 3$?

A. 5

B. 4

C. 3

D. 2

E. 1

35. At the opening of a trading day at a certain stock exchange, the price per share of stock K was \$8. If the price per share of stock K was \$9 at the closing of the day, what was the percent increase in the price per share of stock K for that day?

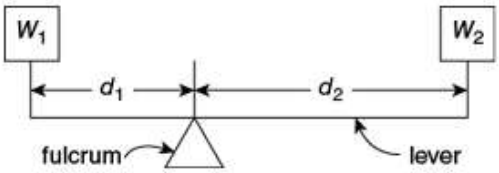
A. 1.4%

B. 5.9%

C. 11.1%

D. 12.5%

E. 23.6%



36. As shown in the diagram above, a lever resting on a fulcrum has weights of w_1 pounds and w_2 pounds, located d_1 feet and d_2 feet from the fulcrum. The lever is balanced and $w_1 d_1 = w_2 d_2$. Suppose w_1 is 50 pounds and w_2 is 30 pounds. If d_1 is 4 feet less than d_2, what is d_2, in feet?

A. 1.5

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. 1.5

B. 2.5

C. 6

D. 10

E. 20

Day	Change in Dollars
Monday	$+1\frac{1}{2}$
Tuesday	$-\frac{3}{4}$
Wednesday	0
Thursday	$-\frac{1}{8}$
Friday	$+2\frac{1}{4}$

37. The table above shows the daily change in the price of a certain stock last week. What was the net change in dollars in the price of the stock for the week?

A. $-4\frac{5}{8}$

B. $-2\frac{7}{8}$

C. $+2\frac{7}{8}$

D. $+3\frac{3}{4}$

E. $-4\frac{5}{8}$

38. Of 120 hotel rooms rented one night, some were suites rented for $115 each and the rest were double rooms rented for $85 each. If the total revenue from the room rentals for this night was $10,890, how many suites were rented?

A. 23

B. 35

C. 54

D. 94

E. 99

39. Last week Jack worked 70 hours and earned $1,260. If he earned his regular hourly wage for the first 40 hours worked, $1\frac{1}{2}$ times his regular hourly wage for the next 20 hours worked, and 2 times his regular hourly wage for the remaining 10 hours worked, what was his regular hourly wage?

A. $7.00

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. $14.00

C. $18.00

D. $22.00

E. $31.50

40. If $\frac{x}{4}$ is 2 more than $\frac{x}{8}$, then $x =$

A. 4

B. 8

C. 16

D. 32

E. 64

41. If $(2^x)(2^y) = 8$ and $(9^x)(3^y) = 81$, then $(x, y) =$

A. (1, 2)

B. (2, 1)

C. (1, 1)

D. (2, 2)

E. (1, 3)

42. A certain toll station on a highway has 7 tollbooths, and each tollbooth collects $0.75 from each vehicle that passes it. From 6 o'clock yesterday morning to 12 o'clock midnight, vehicles passed each of the tollbooths at the average rate of 4 vehicles per minute. Approximately how much money did the toll station collect during that time period?

A. $1,500

B. $3,000

C. $11,500

D. $23,000

E. $30,000

43. If $y = 2 + 2k$ and $y \neq 0$, then $\frac{1}{y} + \frac{1}{y} + \frac{1}{y} + \frac{1}{y} =$

A. $\frac{1}{8+8k}$

B. $\frac{2}{1+k}$

C. $\frac{1}{8+k}$

D. $\frac{4}{8+k}$

E. $\frac{4}{1+k}$

44. If $3^x - 3^{x-1} = 162$, then $x(x-1) =$

A. 12

B. 16

C. 20

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

D. 30

E. 81

45. The expression $n!$ is defined as the product of the integers from 1 through n. If p is the product of the integers from 100 through 299 and q is the product of the integers from 200 through 299, which of the following is equal to $\frac{p}{q}$?

A. 99!

B. 199!

C. $\frac{199!}{99!}$

D. $\frac{299!}{99!}$

E. $\frac{299!}{199!}$

46. A school club plans to package and sell dried fruit to raise money. The club purchased 12 containers of dried fruit, each containing $16\frac{3}{4}$ pounds. What is the maximum number of individual bags of dried fruit, each containing $\frac{1}{4}$ pounds, that can be sold from the dried fruit the club purchased?

A. 50

B. 64

C. 67

D. 768

E. 804

Height	Price
Less than 5 ft	$14.95
5 ft to 6 ft	$17.95
Over 6 ft	$21.95

47. A nursery sells fruit trees priced as shown in the chart above. In its inventory 54 trees are less than 5 feet in height. If the expected revenue from the sale of its entire stock is estimated at $2,450, approximately how much of this will come from the sale of trees that are at least 5 feet tall?

A. $1,730

B. $1,640

C. $1,410

D. $1,080

E. $810

48. A certain bridge is 4,024 feet long. Approximately how many minutes does it take to cross this bridge at a constant speed of 20 miles per hour? (1 mile = 5,280 feet)

A. 1

B. 2

C. 4

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

C. 4

D. 6

E. 7

49. A purse contains 57 coins, all of which are nickels, dimes, or quarters. If the purse contains x dimes and 8 more nickels than dimes, which of the following gives the number of quarters the purse contains in terms of x?

A. $2x - 49$

B. $2x + 49$

C. $2x - 65$

D. $49 - 2x$

E. $65 - 2x$

50. The annual interest rate earned by an investment increased by 10 percent from last year to this year. If the annual interest rate earned by the investment this year was 11 percent, what was the annual interest rate last year?

A. 1%

B. 1.1%

C. 9.1%

D. 10%

E. 10.8%

51. A total of 5 liters of gasoline is to be poured into two empty containers with capacities of 2 liters and 6 liters, respectively, such that both containers will be filled to the same percent of their respective capacities. What amount of gasoline, in liters, must be poured into the 6-liter container?

A. $4\frac{1}{2}$

B. 4

C. $3\frac{3}{4}$

D. 3

E. $1\frac{1}{4}$

52. What is the larger of the 2 solutions of the equation $x^2 - 4x = 96$?

A. 8

B. 12

C. 16

D. 32

E. 100

$$x = \frac{1}{6}gt^2$$

53. In the formula shown, if g is a constant and $x = -6$ when $t = 2$, what is the value of x when $t = 4$?

A. –24

B. –20

C. –15

D. 20

E. 24

(39.897)(0.0096)

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

54. $\frac{(39{,}897)(0.0096)}{198.76}$ is approximately

A. 0.02

B. 0.2

C. 2

D. 20

E. 200

55. The "prime sum" of an integer n greater than 1 is the sum of all the prime factors of n, including repetitions. For example, the prime sum of 12 is 7, since 12 = 2 × 2 × 3 and 2 + 2 + 3 = 7. For which of the following integers is the prime sum greater than 35?

A. 440

B. 512

C. 620

D. 700

E. 750

56. Each machine at a toy factory assembles a certain kind of toy at a constant rate of one toy every 3 minutes. If 40 percent of the machines at the factory are to be replaced by new machines that assemble this kind of toy at a constant rate of one toy every 2 minutes, what will be the percent increase in the number of toys assembled in one hour by all the machines at the factory, working at their constant rates?

A. 20%

B. 25%

C. 30%

D. 40%

E. 50%

57. When a subscription to a new magazine was purchased for m months, the publisher offered a discount of 75 percent off the regular monthly price of the magazine. If the total value of the discount was equivalent to buying the magazine at its regular monthly price for 27 months, what was the value of m?

A. 18

B. 24

C. 30

D. 36

E. 48

58. At a garage sale, all of the prices of the items sold were different. If the price of a radio sold at the garage sale was both the 15th highest price and the 20th lowest price among the prices of the items sold, how many items were sold at the garage sale?

A. 33

B. 34

C. 35

D. 36

E. 37

59. Half of a large pizza is cut into 4 equal-sized pieces, and the other half is cut into 6 equal-sized pieces. If a person were to eat 1 of the larger pieces and 2 of the smaller pieces, what fraction of the pizza would remain uneaten?

A. $\frac{5}{}$

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. $\frac{}{12}$

B. $\frac{13}{24}$

C. $\frac{7}{12}$

D. $\frac{2}{3}$

E. $\frac{17}{24}$

60. If $a = 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}$ and $b = 1 + \frac{1}{4}a$, then what is the value of $a - b$?

A. $-\frac{85}{256}$

B. $-\frac{1}{256}$

C. $-\frac{1}{4}$

D. $\frac{125}{256}$

E. $\frac{169}{256}$

61. In a certain learning experiment, each participant had three trials and was assigned, for each trial, a score of either –2, –1, 0, 1, or 2. The participant's final score consisted of the sum of the first trial score, 2 times the second trial score, and 3 times the third trial score. If Anne received scores of 1 and –1 for her first two trials, not necessarily in that order, which of the following could NOT be her final score?

A. –4

B. –2

C. 1

D. 5

E. 6

62. For all positive integers m and v, the expression $m \Theta v$ represents the remainder when m is divided by v. What is the value of $((98 \Theta 33) \Theta 17) - (98 \Theta (33 \Theta 17))$?

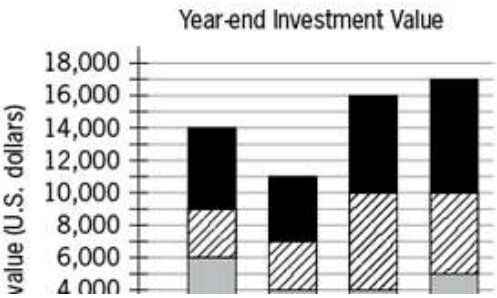
A. −10

B. −2

C. 8

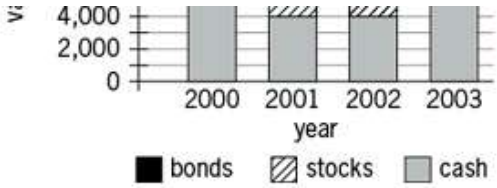
D. 13

E. 17



DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.



63. The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value from year-end 2000 to year-end 2003?

A. $1,000

B. $2,000

C. $3,000

D. $4,000

E. $5,000

64. If the sum of the reciprocals of two consecutive odd integers is $\frac{12}{35}$, then the greater of the two integers is

A. 3

B. 5

C. 7

D. 9

E. 11

65. What is the sum of the odd integers from 35 to 85, inclusive?

A. 1,560

B. 1,500

C. 1,240

D. 1,120

E. 1,100

66. For all numbers a, b, c, and d, $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is defined by the equation $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$. Which of the following is equal to $\begin{vmatrix} s & t \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -t & 2 \\ s & 4 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ t & s \end{vmatrix}$?

A. $\begin{vmatrix} s & t \\ 1 & 5 \end{vmatrix}$

B. $\begin{vmatrix} s & t \\ 7 & 1 \end{vmatrix}$

C. $\begin{vmatrix} s & t \\ 5 & 7 \end{vmatrix}$

D. $\begin{vmatrix} s & -t \\ 1 & 5 \end{vmatrix}$

E. $\begin{vmatrix} s & -t \\ 1 & 7 \end{vmatrix}$

67. In a certain sequence, each term after the first term is one-half the previous term. If the tenth term of the sequence is between 0.0001 and 0.001, then the twelfth term of the sequence is between

A. 0.0025 and 0.025

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. 0.0025 and 0.025

B. 0.00025 and 0.0025

C. 0.000025 and 0.00025

D. 0.0000025 and 0.000025

E. 0.00000025 and 0.0000025

68. A certain drive-in movie theater has a total of 17 rows of parking spaces. There are 20 parking spaces in the first row and 21 parking spaces in the second row. In each subsequent row there are 2 more parking spaces than in the previous row. What is the total number of parking spaces in the movie theater?

A. 412

B. 544

C. 596

D. 632

E. 692

69.

DO NOT COPY
filippo.fiz@uptoten.it